Joint Delay and Doppler Shift Estimation for Multiple Targets Using Exponential Ambiguity Function

Lu Zhang, *Member, IEEE*, Bo Yang, and Maokang Luo

***Abstract*—The estimation of delay and Doppler shift parameters in the presence of multiple targets in radar applications are addressed in this study. The construction of a new ambiguity function, namely, exponential ambiguity function, is proposed by introducing an exponent p into the traditional ambiguity function. Some important properties of the exponential ambiguity function are derived. Then the joint delay and Doppler shift estimation algorithm is proposed based on the combination of the exponential ambiguity function and wavelet denoising method. The performance of the proposed estimation algorithm is theoretically analyzed, verified via Monte Carlo simulation, and compared with the performance of the traditional ambiguity function. Simulation results and theoretical performance analysis show that the proposed estimation algorithm outperforms the existing methods for closely located targets.**

***Index Terms*—Radar signal, joint delay and Doppler shift estimation, resolution, ambiguity function.**

I. Introduction

IN RADAR systems, the estimation of position and velocity plays an important role in the process of locating targets. For slowly fluctuating point targets, position and velocity estimation is accomplished by measuring the differential delay and Doppler shift parameters between the transmitted signal $r(t)$ and the received echo signal $x(t)$, where $x(t)$ is composed of returns from multiple targets and can be written as

$$x(t) = \sum_{l=1}^{L} \alpha_l r(t+\tau_l) e^{-j2\pi\xi_l t} + n(t). \tag{1}$$

In this equation, α_l, τ_l, and ξ_l denote the unknown constant scattering coefficient, delay, and Doppler shift of the l-th target, respectively, $l = 1, 2, ..., L$; and $n(t)$ is an additive white Gaussian noise with zero mean and unknown variance σ^2.

The resolution of a radar system describes the ability of such a system to resolve two or more targets that are closely spaced in range and/or velocity [1]. The resolution is mainly determined by the input signal-to-noise ratio (SNR), radar waveform, and signal processing approach. In a general radar system, the classical ambiguity function (AF) is always used as the typical signal processing approach to estimate the delay and Doppler shift [2], [3]. In this case, for a fixed input SNR, the resolution is only related to the radar waveform; thus, the resolution of such a system is called the inherent resolution in [1]. However, the AF of the transmitted signal is not an ideal single pulse, but it consists of a spikelike mainlobe and numerous noise-like sidelobes. The targets that are extremely close can have overlapping ambiguity functions. This scenario may blur the exact location of a target or make uncertain how many targets are located in a given region in the delay-Doppler shift (τ, ξ) plane [4].

Thus, how to modify the traditional AF to improve the resolution (so as to increase the accuracy of parameter estimation) is an important technical problem to be solved in the presence of multiple targets, particularly when the radar waveform is not ideal or non-cooperative. In the history of AF modification, for the linear frequency modulation signal, scholars proposed many kinds of processing methods, including Wigner Ville transform [5], Radon-Hough transform [6], chirplet transform [7] and fractional Fourier transform [8]. For the polynomial phase signal, the identify ability of multi-component parameter was resolved in [9] using the multilag high-order ambiguity Function. For a general signal, if the targets are stationary, then the traditional approach to improve range resolution is to apply an finite impulse response windowing filter to the received signal to ensure a sharp peak in the obtained cross-correlation [10]–[11]. In fact, sidelobe weighting processing normally widens the mainlobe and leads to SNR loss caused by the mismatch processing. However, only few studies have discussed the problem of joint delay and Doppler shift estimation for multiple moving targets.

In our previous work [12], the exponential filter is proposed as an extension of the classical matched filter for delay estimation of multiple targets problem, which has a sensitivity exponent p that controls its output SNR and range resolution. In this study, we extend the aforementioned exponential filter from one to two dimensions, and a new time-frequency analysis tool, namely, exponential ambiguity function (EAF) with an alterable exponent p, is obtained for joint delay and Doppler shift estimation when the target is detected. The traditional AF is the special EAF with exponent 1. Both the output SNR and the velocity resolution increase as p increases from -1 to 1, whereas the range resolution decreases. Therefore, a reduction of the exponent is needed to improve range resolution; however, this reduction can inevitably degrade the output SNR and velocity resolution. Therefore, we

Manuscript received May 9, 2016; revised November 21, 2016; accepted January 7, 2017. Date of publication January 19, 2017; date of current version February 9, 2017. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Chandra Murthy. This work was supported in part by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 11401405 and Grant 11501386 and in part by the National Natural Science Foundation of China under Grant 11171238.

The authors are with the College of Mathematics, Sichuan University, Chengdu 610065, China (e-mail: zhanglumail@gmail.com; boyang1012@163.com; makaluo@scu.edu.cn).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TSP.2017.2655494

1053-587X © 2017 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

need to combine at least two EAFs with different exponents to achieve the parameter estimation. In addition, if the additive noise is a Gaussian noise, then the output noise of EAF remains a Gaussian noise. The wavelet thresholding method has been used successfully as a useful mathematical tool of denoising in Gaussian noise applications [13], [14], which has a key idea as follows: the wavelet coefficient of the signal after wavelet transform is large and is mainly concentrated in the high frequency area, whereas the wavelet coefficient of the noise is small and disperses in all frequency area. By setting a threshold, the wavelet coefficients greater than the threshold are considered to belong to the useful signal and should be retained. The wavelet coefficients smaller than the threshold are considered to belong to the noise and set to zero so as to achieve the purpose of denoising. We used the wavelet threshold denoising method in [15] and a very good denoising performance was obtained. Therefore, in order to improve the parameter estimation accuracy, we also used the wavelet threshold denoising method in this study to filter the output of the EAF.

The paper is organized as follows. In Section II, we motivate the definitions of EAF and exponential cross ambiguity function (ECAF), after which, the properties of the EAF are derived. Moreover, the properties of the range resolution, velocity resolution, and output noise of the EAF are analyzed, and then the estimation performance of the ECAF is provided by using perturbation analysis. We define the joint exponential cross ambiguity function (JECAF) in Section III. The joint delay and Doppler shift estimation algorithm is proposed based on JECAF and wavelet denoising method. The performance of the proposed estimation algorithm is provided in Section IV, and it is compared with that of the traditional AF. Finally, the conclusions are drawn in Section V.

II. Definition and Properties of the Exponential Ambiguity Function

A. Definition of the Exponential Ambiguity Function

A commonly used tool for estimating delay and Doppler shift is the AF that was introduced by Woodward [2]. The frequency domain expression of the AF of a bandlimited signal $r(t)$ is

$$\chi(\tau,\xi)=\int_{-\infty}^{+\infty}R(f-\xi)R^*(f)e^{-j2\pi f\tau}\mathrm{d}f, \tag{2}$$

where $R(f)=|R(f)|e^{j\theta_R(f)}$ is the Fourier transform of $r(t)$, $|R(f)|$ and $\theta_R(f)$ denote its magnitude and phase respectively, $R^*(f)$ is its complex conjugate.

From the viewpoint of filter theory, for any given ξ, $\chi(\tau,\xi)$ can be seen as the output of a filter with frequency response $R^*(f)$ and input signal $r(t)e^{j2\pi\xi t}$. We rewrite the frequency response $R^*(f)$ in (2) by $|R(f)|e^{-j\theta_R(f)}$, and change the exponent of $|R(f)|$ from 1 to p, and thus obtain the frequency domain expression of the EAF.

Definition 1: The *exponential ambiguity function* (EAF) with exponent p of $r(t)$ is defined by

$$\chi_{rr}^p(\tau,\xi)=\int_{-\infty}^{+\infty}R(f-\xi)R_p^*(f)e^{-j2\pi f\tau}\mathrm{d}f, \tag{3}$$

where $p\in[-1,1]$, and

$$R_p(f)=|R(f)|^p e^{j\theta_R(f)}. \tag{4}$$

Obviously, if $p=1$, $\chi_{rr}^1(\tau,\xi)$ is the classical AF $\chi(\tau,\xi)$ in (2).

Definition 2: The *exponential cross ambiguity function* (ECAF) with exponent p of $r(t)$ and $u(t)$ is defined by

$$\chi_{ru}^p(\tau,\xi)=\int_{-\infty}^{+\infty}U(f-\xi)R_p^*(f)e^{-j2\pi f\tau}\mathrm{d}f, \tag{5}$$

where $R(f)$ and $U(f)$ are the Fourier transforms of $r(t)$ and $u(t)$, respectively. When $p=1$, $\chi_{ru}^1(\tau,\xi)$ is the traditional cross-ambiguity function (CAF) of $r(t)$ and $u(t)$.

Alternative definitions of the EAF can be obtained as follows. Let $F[\cdot]$ and $F^{-1}[\cdot]$ denote the Fourier transform and inverse Fourier transform, respectively. And suppose the inverse Fourier transform of $R_p(f)$ is:

$$r_p(t)=F^{-1}[R_p(f)]. \tag{6}$$

Then, from (3) the Fourier transform of the EAF $\chi_{rr}^p(\tau,\xi)$ can be written by

$$F[\chi_{rr}^p(\tau,\xi)]=F[r(t)e^{j2\pi\xi t}]\cdot F[r_p^*(-t)]. \tag{7}$$

Thus, Let $\otimes$ denote the convolution operator, one has the expression of the EAF of $r(t)$ in time domain

$$\begin{aligned}\chi_{rr}^p(\tau,\xi) &= [r(t)e^{j2\pi\xi t}]\otimes[r_p^*(-t)]\\ &= \int_{-\infty}^{+\infty}r(t)r_p^*(t+\tau)e^{j2\pi\xi t}\mathrm{d}t.\end{aligned} \tag{8}$$

Similarly, the time domain expression of the ECAF of $r(t)$ and $u(t)$ is

$$\chi_{ru}^p(\tau,\xi)=\int_{-\infty}^{+\infty}u(t)r_p^*(t+\tau)e^{j2\pi\xi t}\mathrm{d}t. \tag{9}$$

Supposing the echo signal generated by a single point target is $s(t)=\alpha_0 r(t+\tau_0)e^{-j2\pi\xi_0 t}$, then substituting its Fourier transform $S(f)$ into (5) and using (3) yield

$$\chi_{rs}^p(\tau,\xi)=\alpha_0 e^{-j2\pi(\xi-\xi_0)\tau_0}\chi_{rr}^p(\tau-\tau_0,\xi-\xi_0). \tag{10}$$

One can seen that, if the factor $\alpha_0 e^{-j2\pi(\xi-\xi_0)\tau_0}$ is ignored, the ECAF magnitude is a delayed and Doppler shifted version of the EAF magnitude.

B. Properties of the Exponential Ambiguity Function

As discussed from the previous definitions, EAF is a generalization of the traditional AF. Hence, EAF has many properties similar to those of AF, but the former also has many special properties because of the introduction of the exponent p. The following is a series of the properties of EAF that can provide the basis for parameter estimation.

Property 1: For any $p\in[-1,1]$, $|\chi_{rr}^p(\tau,\xi)|\le|\chi_{rr}^p(0,0)|$. The equality holds if and only if $\tau=0,\xi=0$.

The proof of Property 1 can be completed by reference to the proof of Property 1 in [16]. From Property 1, we can determine that the ECAF of $r(t)$ and $s(t)=\alpha_0 r(t+\tau_0)e^{-j2\pi\xi_0 t}$ satisfies $|\chi_{rs}^p(\tau,\xi)|\le|\alpha_0|\cdot|\chi_{rr}^p(0,0)|$, and the equality holds

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

if and only if $\tau = \tau_0, \xi = \xi_0$. In other words, the returns from the nominal target are located at the origin of the EAF. Therefore, we can estimate the parameters of targets by searching the maximum magnitude points of ECAF. Moreover, the sharper the EAF around $(0,0)$, the better the range and velocity resolution are.

Property 2: Let the p-EAFs of $u(t)$ and $v(t)$ be $\chi^p_{uu}(\tau,\xi)$ and $\chi^p_{vv}(\tau,\xi)$, respectively. Then $\chi^p_{uu}(\tau,\xi)$ is equal to $\chi^p_{vv}(\tau,\xi)$ if and only if $v(t) = cu(t)$ and $|c| = 1$.

The proof of Property 2 is given in Appendix A. This finding shows that for any signal, the following condition applies: if the exponent p is given, then its EAF is unique. Therefore, different signals have different p-EAFs, which also provide a necessary and sufficient condition for waveform design based on EAF.

Property 3: For any $p \in [-1,1]$, the total volume under EAF is constant. In particular,

$$\iint_{-\infty}^{+\infty} |\chi^p_{rr}(\tau,\xi)|^2 \mathrm{d}\tau d\xi = E_1 E_p, \tag{11}$$

where E_p is called the *exponential energy* of $r(t)$, which can be written as

$$E_p = \iint_{-\infty}^{+\infty} |R(f)|^{2p} \mathrm{d}f = \int_{-\infty}^{+\infty} |r_p(t)|^2 \mathrm{d}t. \tag{12}$$

The proof of Property 3 is given in Appendix B. The energy of $r(t)$ is denoted as E, and then $E = E_1$ is obtained from (4) and (12). This finding indicates that the total volume under the EAF surface is only determined by the exponential energy of $r(t)$. In particular, when $p = 1$, Property 3 is the traditional ambiguity principle (i.e., AF has a constant energy). Therefore, we call Property 3 as *exponential ambiguity principle*. Thus, under the constraint of this principle, we can change the waveform of the transmitted signal if the radar waveform is alterable; if the radar waveform is unalterable, then we can also change the exponent of EAF to change the shape of the ambiguity function surface, thereby realizing the matching between the ambiguity function surface and radar target environment.

Property 4:

$$\begin{aligned}\chi_\tau(\xi) &= \int_{-\infty}^{+\infty} |\chi^p_{rr}(\tau,\xi)|^2 \mathrm{d}\tau \\ &= \int_{-\infty}^{+\infty} \chi^1_{rr}(\tau,0)[\chi^1_{r_p r_p}(\tau,0)]^* e^{-j2\pi\xi\tau} \mathrm{d}\tau \\ \chi_\xi(\tau) &= \int_{-\infty}^{+\infty} |\chi^p_{rr}(\tau,\xi)|^2 \mathrm{d}\xi \\ &= \int_{-\infty}^{+\infty} \chi^1_{rr}(0,\xi)[\chi^1_{r_p r_p}(0,\xi)]^* e^{j2\pi\xi\tau} \mathrm{d}\xi\end{aligned}$$

In the above equation, $\chi^1_{rr}(\tau,\xi)$ is the AF of $r(t)$ and $\chi^1_{r_p r_p}(\tau,\xi)$ is the AF of $r_p(t)$.

The proof of Property 4 is given in Appendix C. Property 4 illustrates that the distribution of EAF along the ξ axis, $\chi_\tau(\xi)$, depends only on the autocorrelation function $\chi_{rr}(\tau,0)$ of $r(t)$ and the autocorrelation function $\chi^1_{r_p r_p}(\tau,0)$ of $r_p(t)$, and is equal to the Fourier transform of the product of $\chi^1_{r_p r_p}(\tau,0)$ and $[\chi^1_{r_p r_p}(\tau,0)]^*$. Similarly, the distribution of EAF along the τ axis, $\chi_\xi(\tau)$, is equal to the inverse Fourier transform of the product of $\chi^1_{rr}(0,\xi)$ and $[\chi^1_{r_p r_p}(0,\xi)]^*$.

Property 5: For any $p \in [-1,1]$, $|\chi^p_{rr}(\tau,0)| = |\chi^p_{rr}(-\tau,0)|$.

Property 5 shows that the cut along τ-axis of the magnitude of EAF at $\xi = 0$ is symmetric with respect to the origin; however, if $\xi \neq 0$, only the magnitude of the EAF with exponent 1 (i.e., AF) is symmetric with respect to the origin.

Property 6: If $v(t) = u(at)$, then

$$\chi^p_{vv}(\tau,\xi) = \chi^p_{uu}(a\tau,\xi/a)/|a|.$$

Property 7: If $V(f) = U(af)$, then

$$\chi^p_{vv}(\tau,\xi) = \chi^p_{uu}(\tau/a,a\xi)/|a|.$$

Property 8: If $v(t) = u(t-t_0)e^{j2\pi\xi_0(t-t_0)}$, then

$$\chi^p_{vv}(\tau,\xi) = e^{j2\pi\xi_0(t_0-\tau)}\chi^p_{uu}(\tau,\xi).$$

Property 9: If $V(f) = U(f)e^{j\pi a f^2}$, then

$$\chi^p_{vv}(\tau,\xi) = e^{j\pi a\xi^2}\chi^p_{uu}(\tau+a\xi,\xi).$$

Property 10:

$$\iint_{-\infty}^{+\infty} |\chi^p_{rr}(\tau,\xi)|^2 e^{j2\pi(\tau y-\xi z)} \mathrm{d}\tau \mathrm{d}\xi = \chi^p_{rr}(z,y)[\chi^1_{rr}(z,y)]^*.$$

The proofs of Property 5–9 are clearly established. The proof of Property 10 is given in Appendix D.

In the following, we consider the resolutions of the EAF of $r(t)$. First, the range and velocity joint resolution are considered. The squared magnitude of (3), $|\chi^p_{rr}(\tau,\xi)|^2$, represents the ambiguity surface on the (τ,ξ) plane, which consists of a center peak at $(0,0)$ and sidelobes dispersing on the (τ,ξ) plane. To fully consider the resolution problem involving the interference of both the main peak and the sidelobes, we define the *equivalent ambiguity ellipse area* of the EAF as follows:

$$\Delta_{\tau,\xi}(p) = \iint_{-\infty}^{+\infty} |\chi^p_{rr}(\tau,\xi)|^2 \mathrm{d}\tau \mathrm{d}\xi \Big/ |\chi^p_{rr}(0,0)|^2. \tag{13}$$

The reciprocal $J(p) = 1/\Delta_{\tau,\xi}(p)$ of the equivalent ambiguity area is used to measure the range and velocity joint resolution. In particular, the larger $J(p)$ is, the smaller $\Delta_{\tau,\xi}(p)$ is, then EAF is closer to a pulse signal and the easier to distinguish between two adjacent targets.

Substituting $\tau = 0, \xi = 0$ and (12) into (64) in Appendix B yields,

$$|\chi^p_{\tau,\xi}(0,0)|^2 = (E_{(p+1)/2})^2. \tag{14}$$

Hence, the following can be obtained from (11), (13) and (14):

$$\Delta_{\tau,\xi}(p) = E_1 E_p/(E_{(p+1)/2})^2. \tag{15}$$

In addition, the following can be obtained using the Cauchy integral inequality:

$$\begin{aligned}E_1 E_p &= \int_{-\infty}^{+\infty} |R(f)|^2 \mathrm{d}f \cdot \int_{-\infty}^{+\infty} |R(f)|^{2p} \mathrm{d}f \\ &\geq \left[\int_{-\infty}^{+\infty} |R(f)|^{(p+1)/2} \mathrm{d}f\right]^2 = (E_{(p+1)/2})^2.\end{aligned} \tag{16}$$

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

The equality holds if and only if $p = 1$. Thus, $\Delta_{\tau,\xi}(p) \geq 1$ and $J(p) \leq 1$. Furthermore, we have several conclusions, as presented below.

Property 11: If the EAF of $r(t)$ is $\chi_{rr}^{p}(\tau,\xi)$, then we define $J(p)$ as its *range and velocity joint resolution*, which can be rewritten as:

$$J(p) = (E_{(p+1)/2})^2/(E_1 E_p). \tag{17}$$

Moreover, $J(p)$ is a continuous increasing function for $p \in [-1, 1]$. When $p = 1$, $J(p)$ reaches at its maximum value.

Suppose $g(p) = E_{p/2}$, thus, $10\log_{10}(J(p)) = 10\log_{10}((g(p+1))^2/(g(2)g(2p)))$. Then, from the Section 2 Theorem 1 in [12], we can easily get the proof. Property 11 shows that a certain restriction of the range and velocity joint resolution exists, that is, the equivalent ambiguity area of $r(t)$ is only related to the exponential energy. When $p = 1$, EAF is the traditional AF, which has the largest range and velocity joint resolution and the smallest equivalent ambiguity area.

Next, we consider the range resolution of $\chi_{rr}^{p}(\tau,\xi)$, which depends on the zero-Doppler cut of the EAF, i.e.,

$$\chi_{rr}^{p}(\tau,0) = \int_{-\infty}^{+\infty} R(f)R_p^*(f)e^{-j2\pi f\tau}\mathrm{d}f. \tag{18}$$

Assuming $\xi = 0$ in Property 1, $|\chi_{rr}^{p}(\tau,0)| \leq |\chi_{rr}^{p}(0,0)|$ is obtained. Thus, similar to the AF, the ratio of energy to peak power of $\chi_{rr}^{p}(\tau,0)$, namely, $\sigma_\tau(p)$, is defined as the *equivalent delay width*. The reciprocal of $\sigma_\tau(p)$:

$$T(p) = |\chi_{rr}^{p}(0,0)|^2 \Big/ \int_{-\infty}^{+\infty} |\chi_{rr}^{p}(\tau,0)|^2\mathrm{d}\tau, \tag{19}$$

is then called the *range resolution* of EAF of $r(t)$. The larger the $T(p)$ is, the sharper the function EAF is around $\tau = 0$ in delay dimension, and the easier to distinguish two adjacent targets in delay dimension.

Property 12: For any given $r(t)$, the range resolution of its EAF, namely, $T(p)$, can be simplified as

$$T(p) = (E_{(p+1)/2})^2/E_{p+1}, \tag{20}$$

where $T(p)$ is a continuous function, which is a decreasing function for $p \in [-1, 1]$. When $p = -1$, $T(p)$ reaches at its maximum value.

Similarly, $10\log_{10}(T(p)) = -10\log_{10}(g(2+2p)/g^2(1+p))$, from the Section 2, Theorem 1 in [12], we can easily get the proof. Property 12 shows that the EAF with $p = 1$ is the traditional AF, which has the smallest (worst) range resolution. The range resolution increases as the exponent p decreases at $[-1, 1]$. In other words, the EAF with $p < 1$ presents a narrower mainlobe and lower side lobe in delay dimension when compared with the classical AF. Therefore, choosing a small p can improve the range resolution of EAF.

Similarly, supposing the *equivalent frequency bandwidth* of EAF along the ξ axis is $\sigma_\xi(p)$, then we have,

$$\sigma_\xi(p) = \int_{-\infty}^{+\infty} |\chi_{rr}^{p}(0,\xi)|^2\mathrm{d}\xi \Big/ |\chi_{rr}^{p}(0,0)|^2, \tag{21}$$

where $\chi_{rr}^{p}(0,\xi)$ is the zero-range cut of EAF. Notice that the cross-section of the EAF can be approximately regarded as an ellipse, one has $\sigma_\tau(p)\sigma_\xi(p)$ is in proportion to $\Delta_{\tau,\xi}(p)$. Since $\sigma_\tau(p)$ is an increasing function of p and $\Delta_{\tau,\xi}(p)$ is a decreasing function of p, then $\sigma_\xi(p)$ is a decreasing function of p. Similarly, we define $D(p) = 1/\sigma_\xi(p)$ as the *velocity resolution*, where $D(p)$ is also a continuous function of p, which decreases for $p \in [-1, 1]$. When $p = 1$, $D(p)$ achieves its maximum value, then the AF has the highest (best) velocity resolution. In other words, the EAF with $p < 1$ shows a wider mainlobe and higher side lobe in Doppler shift dimension when compared with the AF.

Finally, as an example, Property 13 gives the analytical expression for EAF of a signal whose complex envelope is a simple Gaussian pulse.

Property 13: The Gaussian pulse $u(t) = \exp(-t^2/(2a^2))$ has EAF given by

$$\chi_{uu}^{p}(\tau,\xi) = \frac{(2\pi a^2)^{\frac{p+1}{2}}}{\sqrt{2a(p+1)}}\exp\left\{-\frac{2\pi}{p+1} \times \left(a^2 p\xi^2 + j\xi\tau + \frac{\tau^2}{4\pi a^2}\right)\right\}. \tag{22}$$

From the pair of Fourier transform $e^{-\pi t^2} \leftrightarrow e^{-\pi f^2}$, we can easily obtain the proof. Moreover, the square of (22) with $\tau = 0$ and $\xi = 0$ are

$$|\chi_{uu}^{p}(0,\xi)|^2 = \frac{(a\sqrt{2\pi})^p}{\sqrt{p+1}}\exp\left\{-\frac{4\pi a^2 p}{p+1}\xi^2\right\}, \tag{23}$$

$$|\chi_{uu}^{p}(\tau,0)|^2 = \frac{(a\sqrt{2\pi})^p}{\sqrt{p+1}}\exp\left\{-\frac{1}{a^2(p+1)}\tau^2\right\}. \tag{24}$$

From (23) and (24), we can see, for a Gaussian pulse, if $p \in (0, 1]$, the cut of its EAF along τ axis or ξ axis are still Gaussian signals; if $p \in [-1, 0]$, the cut of its EAF along τ axis is still a Gaussian signal while the cut along ξ axis is an exponential signal. Moreover, the width of $u(t) = \exp(-t^2/(2a^2))$ is $a\sqrt{2\pi}$. Therefore, for any given $p > 0$, EAF of the Gaussian pulse has a width $\sigma_\tau(p)$ along τ axis and a width $\sigma_\xi(p)$ along ξ axis, which can be written as

$$\sigma_\tau(p) = a\sqrt{\pi(p+1)},\ \sigma_\xi(p) = \frac{1}{2a}\sqrt{1+\frac{1}{p}}. \tag{25}$$

Thus, for $p > 0$, the EAF of this kind of signal seems like a *ridge* in the (τ,ξ) plane. Let $p = 1$ in (25), then the classical AF has a time-width $\sigma_\tau(1) = a\sqrt{2\pi}$ and a frequency-width $\sigma_\xi(1) = 1/(\sqrt{2}a)$. Therefore, EAF can be obtained by elongating AF $\sqrt{(p+1)/2}$ times along the τ axis and $\sqrt{(p+1)/(2p)}$ times along the ξ axis. Fig. 1 shows the normalized magnitude plots and equal-height contour plots (-3 dB cut) of the AF and EAF ($p = 0.1$) of a Gaussian pulse with $a = 2$.

As another example, we consider a frequency-stepped pulse signal has a form of

$$r(t) = \sum_{n=1}^{N} \mathrm{rect}(t - nT_s)\cos\big(2\pi(f_0 + n\Delta f)t\big), \tag{26}$$

in which, $\mathrm{rect}(t)$ represents a rectangle signal, $N = 20$, $T_s = 0.3$ s, $f_0 = 5$ Hz and $\Delta f = 3$ Hz. We plot the normalized magnitudes of the classical AF, EAF with $p = 0$ of this signal and their

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

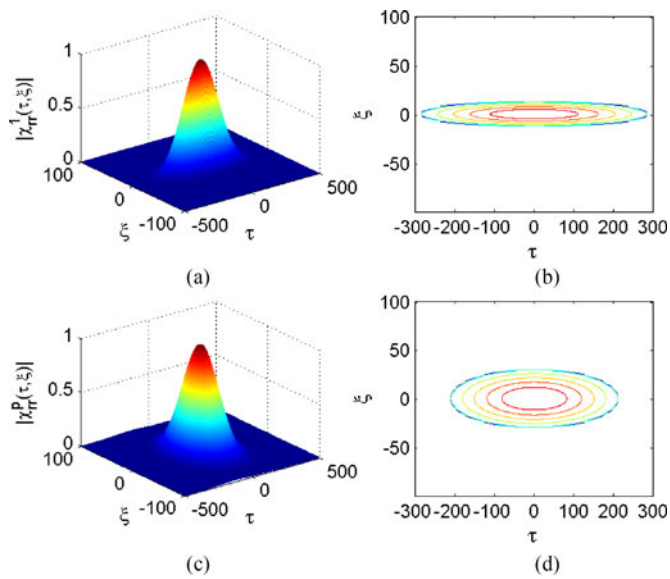


Fig. 1. The plots of the AF and EAF ($p = 0.1$) of a Gaussian pulse with $a = 2$. (a) Normalized magnitude of AF. (b) Contour plot of AF. (c) Normalized magnitude of EAF, $p = 0.1$. (d) Contour plot of EAF, $p = 0.1$.

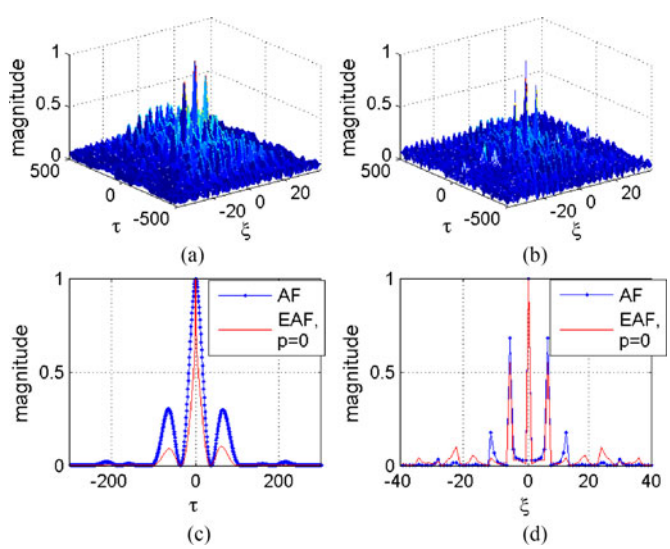


Fig. 2. The plots of AF, EAF with $p = 0$ of a frequency-stepped pulse signal. (a) Normalized magnitude of AF. (b) Normalized magnitude of EAF. (c) Zero-Doppler cut. (d) Zero-range cut.

zero-Doppler cuts, zero-range cuts in Fig. 2. We can observe that, when compared to AF, the EAF of this frequency-stepped pulse signal has a narrower main lobe and a lower sidelobe energy in τ dimension, while has a slightly wider main lobe and a slightly higher sidelobe energy in ξ dimension.

C. Properties of Output Noise and Estimation Error Analysis

A noisy received signal from a single target can be written as

$$x(t) = \alpha_0 r(t+\tau_0)e^{-j2\pi\xi_0 t} + n(t), \tag{27}$$

where $r(t)$ is the transmitted signal; the parameters α_0, τ_0, and ξ_0 denote the unknown constant scattering coefficient, delay, and Doppler shift, respectively; $n(t)$ is additive white Gaussian noise with zero mean and unknown variance σ^2. The ECAF of $r(t)$ and $x(t)$ is given by

$$\chi_{rx}^p(\tau,\xi) = \alpha_0 e^{-j2\pi(\xi-\xi_0)\tau_0}\chi_{rr}^p(\tau-\tau_0,\xi-\xi_0) + \chi_{rn}^p(\tau,\xi), \tag{28}$$

where the signal function and noise function are given respectively by

$$\chi_{rr}^p(\tau,\xi) = \int_{-\infty}^{+\infty} r(t)r_p^*(t+\tau)e^{j2\pi\xi t}\mathrm{d}t, \tag{29}$$

$$\chi_{rn}^p(\tau,\xi) = \int_{-\infty}^{+\infty} n(t)r_p^*(t+\tau)e^{j2\pi\xi t}\mathrm{d}t. \tag{30}$$

Given that $n(t)$ is Gaussian noise, the weighted integral result of $n(t)$ is still Gaussian noise. Hence, for a given ξ, the ECAF can be seen as a linear filter, and thus, for any τ, the output $\chi_{rn}^p(\tau,\xi)$ is a Gaussian random variable.

By using $E\{n(t)\} = 0$, the mean value of $\chi_{rn}^p(\tau,\xi)$ is expressed as

$$E\{\chi_{rn}^p(\tau,\xi)\} = \int_{-\infty}^{+\infty} E\{n(t)\}r_p^*(t+\tau)e^{j2\pi\xi t}\mathrm{d}t = 0. \tag{31}$$

By using Parseval's formula and (4), (6), (12), its variance can be written as

$$N(p) = E\{\chi_{rn}(\tau,\xi)\chi_{rn}^*(\tau,\xi)\} = \sigma^2 E_p. \tag{32}$$

Here, we use the ratio of maximum output power of the signal to the output noise variance to describe the output SNR of EAF, which is given by

$$\mathrm{SNR}_{out}(p) = \max_{\tau,\xi} \frac{|\alpha_0 e^{-j2\pi(\xi-\xi_0)\tau_0}\chi_{rr}^p(\tau-\tau_0,\xi-\xi_0)|^2}{N(p)}. \tag{33}$$

By Property 1, (14), (17) and (33), the above formula can be simplified as

$$\mathrm{SNR}_{out}(p) = \alpha_0^2(E_{(p+1)/2})^2/(\sigma^2 E_p) = \alpha_0^2 E_1 J(p)/\sigma^2, \tag{34}$$

where $J(p)$ is the range-velocity joint resolution given by (17). Thus, the output SNR of $p-$ EAF clearly has the same monotonicity as $J(p)$. Therefore, from Property 12, we obtain the following property.

Property 14: The output noise function $\chi_{rn}^p(\tau,\xi)$ of the EAF of the noisy received signal $x(t)$ and transmitted signal $r(t)$ is a Gaussian noise with zero mean and variance $N(p) = \sigma^2 E_p$. Moveover, the output SNR, $\mathrm{SNR}_{out}(p)$, is a continuous and increasing function of p in $[-1, 1]$, which reaches its maximum at $p = 1$.

In the following section, since α_0^2/σ^2 in (34) is a fixed constant coefficient, we ignore this coefficient and use the logarithmic output SNR for calculation which is given by

$$\begin{aligned}\lg \mathrm{SNR}_{out}(p) &= 10\lg[\sigma^2\mathrm{SNR}_{out}(p)/\alpha_0^2] \\ &= 10\lg[(E_{(p+1)/2})^2/E_p].\end{aligned} \tag{35}$$

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

D. Error Analysis of the Joint Delay and Doppler Shift Estimation Based on the Exponential Ambiguity Function

Suppose the received signal $x(t)$ from a single target is given by (27), from the above discussion, we know the estimation of delay and Doppler shift can be obtained by finding maximum modulus points of the ECAF, i.e.,

$$\hat{\tau}_0, \hat{\xi}_0 = \arg\max_{\tau,\xi} |\chi_{rx}^p(\tau,\xi)|, \tag{36}$$

where $\chi_{rx}^p(\tau,\xi)$ is given in (28). Given that the extreme value point of a function is equal to its zero point, (36) can thus be replaced by the estimation formula given by

$$\left.\frac{\partial|\chi_{rx}^p(\tau,\xi)|^2}{\partial\tau}\right|_{\hat{\tau}_0,\hat{\xi}_0} = 0, \quad \left.\frac{\partial|\chi_{rx}^p(\tau,\xi)|^2}{\partial\xi}\right|_{\hat{\tau}_0,\hat{\xi}_0} = 0. \tag{37}$$

From Property 14 we know, $\chi_{rx}^p(\tau,\xi)$ is a Gaussian random variable with mean $\alpha_0 e^{-j2\pi(\xi-\xi_0)\tau_0}\chi_{rr}^p(\tau-\tau_0,\xi-\xi_0)$ and variance $\sigma^2 E_p$. Accordingly, owing to the influence of noise, the peak of $|\chi_{rx}^p(\tau,\xi)|^2$ will move from (τ_0,ξ_0) to $(\hat{\tau}_0,\hat{\xi}_0)$. The expansion of the square of (28) is:

$$[\chi_{rx}^p(\tau,\xi)]^2 = |\alpha_0|^2|\mathbf{X}_p|^2 + 2\Re(\alpha_0\mathbf{X}_p\mathbf{N}_p^*) + |\mathbf{N}_p|^2, \tag{38}$$

where $\mathbf{X}_p = e^{-j2\pi(\xi-\xi_0)\tau_0}\chi_{rr}^p(\tau-\tau_0,\xi-\xi_0)$, $\mathbf{N}_p = \chi_{rn}^p(\tau,\xi)$, and $\Re(\cdot)$ denotes the real part.

As we are considering the case of large input SNR, the third item may be ignored when compared to the first and second items. Thus, $[\chi_{rr}^p(\tau,\xi)]^2$ in (38) is approximately equal to $|\alpha_0|^2|\mathbf{X}_p|^2 + 2\Re(\alpha_0\mathbf{X}_p\mathbf{N}_p^*)$, substituting this formula into (37), and using the Taylor linear approximation, we have

$$\begin{aligned}
&\frac{|\alpha_0|^2}{2}\left[\frac{\partial^2|\mathbf{X}_p|^2}{\partial\tau^2}\right]\Bigg|_{\tau_0,\xi_0}(\hat{\tau}_0-\tau_0) + \frac{|\alpha_0|^2}{2}\left[\frac{\partial^2|\mathbf{X}_p|^2}{\partial\tau\partial\xi}\right]\Bigg|_{\tau_0,\xi_0}(\hat{\xi}_0-\xi_0) \\
&\qquad = -\Re\left[\alpha_0\frac{\partial[\mathbf{X}_p\mathbf{N}_p^*]}{\partial\tau}\right]\Bigg|_{\tau_0,\xi_0} \\
&\frac{|\alpha_0|^2}{2}\left[\frac{\partial^2|\mathbf{X}_p|^2}{\partial\tau\partial\xi}\right]\Bigg|_{\tau_0,\xi_0}(\hat{\tau}_0-\tau_0) + \frac{|\alpha_0|^2}{2}\left[\frac{\partial^2|\mathbf{X}_p|^2}{\partial\xi^2}\right]\Bigg|_{\tau_0,\xi_0}(\hat{\xi}_0-\xi_0) \\
&\qquad = -\Re\left[\alpha_0\frac{\partial[\mathbf{X}_p\mathbf{N}_p^*]}{\partial\tau}\right]\Bigg|_{\tau_0,\xi_0}
\end{aligned} \tag{39}$$

Taking the expectation of both sides of and using the properties of the noise function, we obtain

$$E\{\hat{\tau}_0\} = \tau_0, \; E\{\hat{\xi}_0\} = \xi_0. \tag{40}$$

Therefore, the joint delay and Doppler shift estimation for a single target based on ECAF are unbiased. From (39) we know, the root mean square errors (RMSE) of the delay and Doppler shift estimation are not only related to input SNR, but to the parameters of the exponential waveform of the signal.

Given the difficulty of obtaining a closed-form expression for RMSE, we evaluated it numerically and compared the RMSE of ECAF with those of the traditional CAF. In the following simulations, the transmitted signal is chosen as an FM signal with bandwidth 50 KHz, sampling interval $\Delta t = 10^{-6}/3$ s, sampling length $K = 2000\Delta t$, and expression

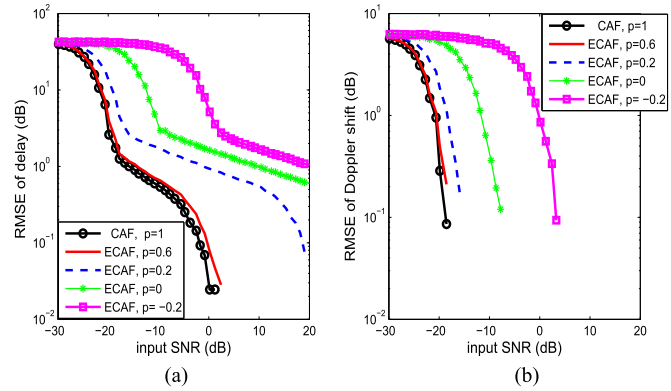


Fig. 3. Estimation error in the case of single target. (a) RMSE of delay. (b) RMSE of Doppler shift.

$$r(t) = \cos(2\pi f_0 t + \int_0^t a(u)\mathrm{d}(u)), \tag{41}$$

where f_0 is the radar carrier frequency and $a(t)$ is the modulating signal. For different input SNR levels, we obtained various p-ECAFs with different exponents, and then searched their maximum modulus points to calculate the RMSE of corresponding delay estimated value $\hat{\tau}_0$ and Doppler shift estimated value $\hat{\xi}_0$.

1) Single target case: First, we assumed that the echo signal contained only a single target. The true parameter values were $(\alpha_0,\tau_0,\xi_0) = (1,20,1)$, in the units of the sampling interval. Fig. 3(a) presents the delay RMSEs in logarithmic axis versus the input SNR of ECAF with different exponents. Fig. 3(b) gives the corresponding Doppler shift RMSEs in logarithmic axis. Among them, the exponents chosen were 1, 0.6, 0.2, 0 and -0.2, of which, the ECAF with exponent 1 was the classical CAF. The range of the input SNR was -30 dB $\sim$20 dB. For each input SNR, we conducted 500 experiments and averaged the results. In these figures, the curves end when the RMSEs are zero because of the logarithm operation, and it is the same to the following RMSE figures that are in logarithmic axis.

As shown in Fig. 3, the delay and Doppler shift RMSEs based on traditional CAF and ECAF decrease with the increase of input SNR. For the same input SNR, the delay RMSEs of ECAF reduce gradually as the exponent p increases. If $p = 1$(as in the case of CAF), the delay and Doppler shift RMSEs reach minimum value, while a loss of output SNR for $p < 1$ is observed, thus resulting in greater estimation error. This is consistent with the conclusion in Property 14. Moreover, for all input SNR, the RMSEs of the ECAF with $p = 0.6$ are quite close to those of the CAF.

2) Multiple targets case: Next, we considerd the case of multiple targets. For simplicity, we assumed that there were only two adjacent targets. The true values of parameters were $(\alpha_1,\tau_1,\xi_1) = (1,131,1), (\alpha_2,\tau_2,\xi_2) = (0.8,161,1)$, in the units of the sampling interval. We used the first target as the main target as it had a greater ambiguity function peak. For ECAF with different exponents, we plot RMSE of delay

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

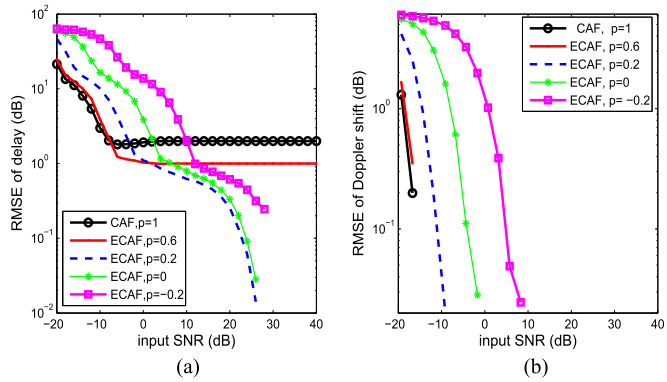


Fig. 4. Estimation error of the main targets in the case of two adjacent targets. (a) RMSE of delay. (b) RMSE of Doppler shift.

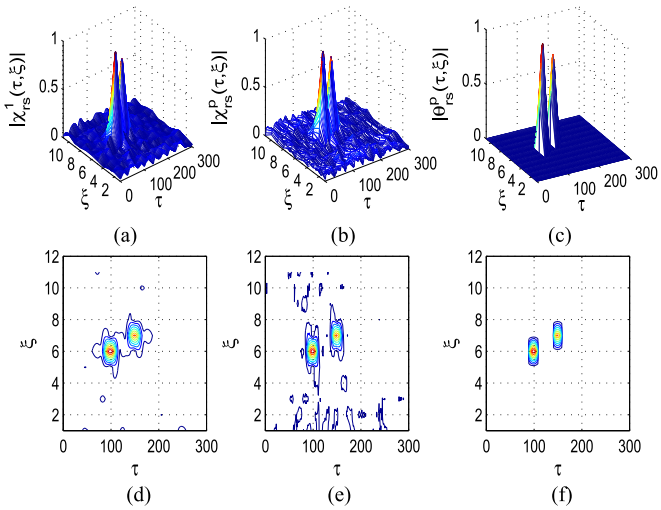


Fig. 5. Normalized magnitude and contour plots of classical CAF, ECAF and JECAF. (a) Normalized magnitude of CAF. (b) Normalized magnitude of ECAF. (c) Normalized magnitude of JECAF. (d) Contour plot of CAF. (e) Contour plot of ECAF. (f) Contour plot of JECAF.

estimation in logarithmic axis in Fig. 4(a), and the Doppler shift RMSE in logarithmic axis in Fig. 4(b), in which, the exponents were chosen as 1, 0.6, 0.2, 0, -0.2. The range of the input SNR was -20 dB $\sim$40 dB. For each input SNR, we conducted 500 experiments and averaged the results.

Fig. 4(a) shows that in the presence of two adjacent targets in the same area, the delay estimation error of the main target based on traditional CAF decreases with the increase of input SNR, and finally tends to a constant much larger than zero. This is because the range resolution of the traditional CAF of the given transmitted signal is a constant, and not high enough to distinguish the two adjacent targets in this scenario. The delay estimation error of the main target, based on ECAF, also decreases as input SNR increases. For low input SNR, the errors of ECAF are larger than those of CAF, and the error of ECAF with exponent $p = 0.6$ is very close to that of traditional CAF. As input SNR increases, the errors of ECAF are smaller than those of CAF. This is because, as the exponent decreases, the range resolution of ECAF gradually increases and thus the ECAF is capable of distinguishing multiple targets. However, ECAF with a smaller exponent does not always have a smaller estimation error. For example, in the whole SNR region, ECAF with exponent $p = 0.2$ has a smaller error than ECAF with exponent $p = 0, -0.2$, because in such a scenario, the range resolution of ECAF with exponent $p = 0.2$ is sufficient to distinguish between these two targets. In case of multiple targets, there is an optimal exponent for the given input SNR, which is used to obtain the minimum delay estimation error. Fig. 4(b) shows that when two adjacent targets exist in the same area, Doppler shift estimation error of the main target, based on traditional CAF, decreases as input SNR increases, finally tending toward 0. The Doppler shift estimation error of the main target based on ECAF also decreases as input SNR increases, finally approaching the estimation error of traditional CAF.

III. Joint Delay and Doppler Shift Estimation Algorithm

If EAF of the transmitted signal $r(t)$ is $\chi^p_{rr}(\tau,\xi)$, then from Property 12 and Property 13, we know that as p decreases in $[-1, 1]$, the range resolution of $\chi^p_{rr}(\tau,\xi)$ increases, and velocity resolution decreases. Hence, using of a single ECAF with fixed exponent to obtain optimal resolution of both range and velocity becomes difficult. To do this, two p-ECAFs must be combined with different exponents to estimate the parameters for ideal range and velocity resolution. One exponent is fixed as 1, because the classical CAF can obtain optimal velocity resolution, while the other exponent p must be determined.

For any given $r(t)$ and $s(t)$, we normalize the magnitude of their ECAF $\chi^p_{rs}(\tau,\xi)$ as,

$$\tilde{\chi}^p_{rs}(\tau,\xi) = |\chi^p_{rs}(\tau,\xi)| / \max_{\tau,\xi} |\chi^p_{rs}(\tau,\xi)|. \tag{42}$$

Assuming the height of the highest side lobe of the normalization function $\tilde{\chi}^p_{rs}(\tau,\xi)$ is $H \in [0, 1]$, the area

$$\mathbf{O}^p_{rr}(H) = \{(\tau,\xi) | \tilde{\chi}^p_{rs}(\tau,\xi) > H\}, \tag{43}$$

then represents the approximate *ambiguity ellipse region*, which is the projection of the cross-section of the center peak of $\tilde{\chi}^p_{rs}(\tau,\xi)$ at height H. Given that the delay and Doppler shift of the targets are concentrated at the peak of the ECAF, they should appear above, specifically in the approximate ambiguity ellipse region. Therefore, we define the *joint exponential cross ambiguity function* (JECAF) with exponent p as

$$\theta^p_{rs}(\tau,\xi) = \begin{cases} \tilde{\chi}^p_{rs}(\tau,\xi), & (\tau,\xi) \in \mathbf{O}^p_{rs}(H) \bigcap \mathbf{O}^1_{rs}(H); \\ 0, & \text{otherwise.} \end{cases} \tag{44}$$

We select the part of ECAF at the intersection of the ambiguity ellipse region of ECAF and the ambiguity ellipse region of CAF for processing. Parts outside the region are set to zero (because they display sidelobes).

As an example, we took the FM signal given by (41) as the transmitted signal, while the noisy received signal of two adjacent targets was written as (1). With input SNR 8 dB, the true values of parameters were set as $(1, 100, 6)$, $(0.8, 150, 7)$, in the units of the sampling interval. Fig. 5 shows the magnitude and

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

contour maps of classical CAF, ECAF with $p = 0$ and JECAF with $p = 0$, in which $H = 0.35$ (H is selected as the height of the highest sidelobe of the normalization of AF). Note that JECAF has three significant advantages as described below.

1) Only two main peaks can be found in the contour plot of JECAF, and the values of (τ, ξ) at the main peak are the delay and Doppler shift of the targets, respectively. Hence, multiple targets detection and parameter estimation can be achieved simultaneously by searching the maximum modulus points of the JECAF.

2) At the dimension of delay, ECAF with $p = 0$ has a smaller equivalent delay width than conventional CAF, and JECAF with $p = 0$ retains this characteristic, thus obtaining a higher range resolution than traditional CAF.

3) At the dimension of Doppler shift, from Property 3, the energy of EAF with $p < 1$ spreads from the delay dimension to the Doppler shift dimension; therefore, it has a higher sidelobe than classical AF. Moreover, JECAF with $p = 0$ retains better performance of low Doppler shift sidelobes when compared to CAF due to the sidelobes suppression effect of the latter.

A. Optimal Exponent Selection

The exponent we propose for JECAF can be chosen arbitrarily in $[-1, 1]$, so we must determine and select the optimal exponent according to a certain rule. From Property 14, we know that for $p \in [-1, 1)$, ECAF has an output SNR loss compared to that of classical CAF. However, the increase of the range resolution is gained at the expense of SNR loss. The smaller the exponent p, the higher the range resolution is, but the lower the output SNR is. Therefore, the choice of exponent can ultimately affect the accuracy of parameter estimation.

Suppose $\lg \mathrm{SNR}_{out}(1)$ is the maximum output SNR obtained by the AF, and from (35) we know, $\lg \mathrm{SNR}_{out}(1) = 10 \lg(E)$, where E is the energy of the transmitted signal $r(t)$. Then for a given SNR loss S_0, the output SNR of EAF, namely, $\lg \mathrm{SNR}_{out}(p)$, must be higher than threshold $S = \lg \mathrm{SNR}_{out}(1)(1 - S_0)$ to ensure adequate output SNR. We then achieve optimal resolution by choosing the exponent. Therefore, the exponent p should satisfy

$$\lg \mathrm{SNR}_{out}(p) \geq S = \lg \mathrm{SNR}_{out}(1)(1 - S_0), \tag{45}$$

where $\lg \mathrm{SNR}_{out}(p)$ is given in (35). Given that $\lg \mathrm{SNR}_{out}(p)$ is an increasing function for $p \in [-1, 1]$, then the above problem (45) is equivalent to $p \geq p_0$, in which p_0 is the solution of $\lg \mathrm{SNR}_{out}(p) = S$, and can be obtained by solving the optimization problem given by

$$p_0 = \arg \min_{p \in [-1,1]} (\lg \mathrm{SNR}_{out}(p) - S)^2. \tag{46}$$

Conversely, we set the target with the highest ambiguity function peak as the main target. If we obtain the estimated values of the main target by searching the maximum modulus points of JECAF with exponent p, i.e., $\hat{\alpha}_1^p, \hat{\tau}_1^p, \hat{\xi}_1^p$, we can calculate the estimated echo of the main target $\hat{s}_1^p(t) = \hat{\alpha}_1^p r(t + \hat{\tau}_1^p) e^{-j2\pi \hat{\xi}_1^p t}$. Thus, subtracting the estimated echo of the main target from the expression of the received signal $x(t)$ in (1) should remove the effects of the main target from the received signal, and the energy of the remaining received signal should be greatly reduced. Therefore, the optimal exponent p_{opt} should minimize the residual energy of the remaining received signal given by

$$p_{opt} = \arg \min_{p \in [p_0, 1]} \int_0^T |x(t) - \hat{s}_1^p(t)|^2 \mathrm{d}t, \tag{47}$$

where, T is the sampling length of $x(t)$.

B. Joint Delay and Doppler Shift Estimation Algorithm Based on the Joint Exponential Ambiguity Function

In this section, we provide the algorithm for estimating parameters of multiple targets. The definition and properties of EAF discussed in the previous sections are all continuous functions on an infinite time interval. However, in practical applications, we process discrete signals with finite length. By the definition (9), we obtain the discrete forms of ECAF as follows

$$\mathbf{X}_{rx}^p[m, n] = \Delta t \sum_{k=0}^{K-1} x[k] \cdot r_p^*[k + m] e^{j2\pi mn}, \tag{48}$$

$0 \leq m \leq M, 0 \leq n \leq N$. Δt is the time sampling interval. Here, FFT and IFFT respectively represent the fast discrete Fourier transform and fast discrete inverse Fourier transform, then from (4) and (6) we have

$$r_p[k] = \mathrm{IFFT}\{|\mathrm{FFT}\{r[k]\}|^{p-1} \cdot \mathrm{FFT}\{r[k]\}\}, \tag{49}$$

$k = 0, 1, ..., K - 1$.

The discrete form of exponential energy given by (12) can be obtained by using

$$E_p = \Delta t \sum_{k=0}^{K-1} (r_p[k])^2. \tag{50}$$

In addition, from Property 14, the output noise of the ECAF of the transmitted signal and noisy received signal, $\chi_{rn}^p(\tau, \xi)$, for a given (τ, ξ), is a Gaussian noise with zero mean and variance $N(p) = E_p \sigma^2$, which greatly affects the final parameter estimation of low input SNR. Thus, we introduced Donoho's wavelet threshold denoising method in [13], [15] into our estimation algorithm to filter the output noise. For each $n = 1, 2, .., N$, the discrete signal $\chi_{rx}^p[m, n], m = 1, 2, ..., M$ is treated as a one-dimensional function of the delay. Let $\mathbf{W}$ be a left invertible wavelet transformation matrix of the discrete wavelet transform (DWF), we propose the following wavelet threshold denoising scheme to filter $\chi_{rx}^p[m, n], m = 1, 2, ..., M$:

(1) Compute the DWT of $\chi_{rx}^p[m, n]$, obtaining the wavelet coefficients $\chi_B[m] = \mathbf{W} \cdot \chi_{rx}^p[m, n], m = 1, 2, ..., M$;

(2) Perform thresholding to the wavelet coefficients χ_B in the wavelet domain, according to soft thresholding:

$$\rho_T(\chi_B) = \begin{cases} \chi_B - T_w, & \text{if } \chi_B > T_w; \\ 0, & \text{if } |\chi_B| \leq T_w \\ \chi_B + T_w, & \text{if } \chi_B < -T_w \end{cases} \tag{51}$$

with the threshold T_w obtained by the method proposed in [13];

(3) Compute the inverse DWT, obtaining the estimated results after denoising as $w\chi_{rx}^p[m, n] = \mathbf{W}^{-1} \rho_T(\chi_B)$.

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

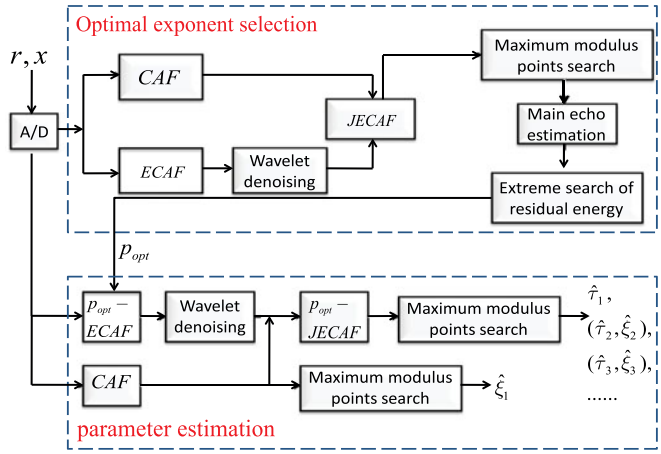


Fig. 6. Block diagram of the proposed joint delay and Doppler shift estimation algorithm.

Supposing the noisy received signal $x(t)$ from L targets is given by (1), the JECAF after wavelet denoising is

$$\theta_{rx}^{p}[m,n] = \begin{cases} w\chi_{rx}^{p}[m,n], & [m,n] \in \mathbf{O}_{rx}^{p}(H) \bigcap \mathbf{O}_{rx}^{1}(H); \\ 0, & \text{otherwise.} \end{cases} \tag{52}$$

We assume that the first target target$_1$ with the highest peak in ambiguity function is the main target. As the traditional CAF has optimal velocity resolution, we adopt the Doppler shift estimation result of the main target based on CAF as the final Doppler shift estimation result, and adopt the delay estimation result of main target based on JECAF with the optimal exponent as the final delay estimation result, in which the optimal exponent is chosen using the method stated in Section III-B. For other targets, the delay and Doppler shift estimation results of the targets based on JECAF with the optimal exponent will be chosen as the final estimation results.

In summary, our joint delay and Doppler shift estimation algorithm mainly consists of the following steps, whose block diagram is shown in Fig. 6.

Step 1: Convert the transmitted signal $r(t)$ and received signal $x(t)$ into the digital signals $r[k]$ and $x[k]$, respectively, $k = 0, 1, ..., K-1$; and get the K-point FFT of $r(t)$ and $x(t)$ as $R[j]$ and $X[j]$, $j = 1, 2, ..., K$; and then obtain $r_p[i], i = 1, 2, ..., K$ by (49);

Step 2: Obtain the discrete form $\mathbf{X}_{rr}^{1}$ of the AF of $r(t)$, the discrete form $\mathbf{X}_{rx}^{1}$ of the CAF of $r(t)$ and $x(t)$; and searching the maximum modulus point of $\mathbf{X}_{rx}^{1}$ to get the delay and Doppler shift estimated values of the main target:

$$\hat{m}, \hat{n} = \arg\max_{m,n} |\mathbf{X}_{rx}^{1}[m,n]|, \tag{53}$$

the corresponding delay and Doppler shift estimated value of CAF are $\hat{\tau}_{caf} = \Delta t \times \hat{m}, \hat{\xi}_{caf} = \Delta\omega \times \hat{n}$.

Step 3: According to the pre-given acceptable SNR loss S_0, obtain the searching lower bound p_0 of the exponent by solving problem (46), and then obtain the initial value $\alpha_{initial}$ of α_1 by using the relationship formula $\max_{\tau,\xi} |\chi_{rx}^{p}(\tau,\xi)|^2 \approx |\alpha_1|^2 |\chi_{rr}^{p}(0,0)|^2$:

$$\alpha_{initial} = \max_{m,n} |\mathbf{X}_{rx}^{1}| / \max_{m,n} |\mathbf{X}_{rr}^{1}|. \tag{54}$$

Insert I points in $[p_0, 1]$, for $p_i = p_0 + i(1-p_0)/I, i = 1, 2, ..., I$, carry out the following steps respectively:

1) Obtain the discrete form of the ECAF of $r(t)$ and $x(t)$ with exponent p_i as $\mathbf{X}_{rx}^{p_i}$;

2) Filter $\mathbf{X}_{rx}^{p_i}$ using the wavelet denoising method and obtain the estimated result $w\mathbf{X}_{rx}^{p_i}$;

3) According to the pre-given threshold value H, the discrete form of the JECAF $\theta_{rx}^{p_i}$ with exponent p_i is obtained by (52), and then searching the maximum points of $\theta_{rx}^{p_i}$ gives the delay and Doppler shift estimated value of the main target:

$$\hat{m}_{p_i}, \hat{n}_{p_i} = \arg\max_{m,n} |\theta_{rx}^{p_i}[m,n]|, \tag{55}$$

the corresponding delay and Doppler shift estimated value of JECAF are $\hat{\tau}_1^{p_i} = \Delta t \times \hat{m}_{p_i}, \hat{\xi}_1^{p_i} = \Delta\omega \times \hat{n}_{p_i}$ and thus we obtain the estimated value of the echo signal of the main target $\hat{s}_1^{i}(t) = \alpha_{initial} r(t + \hat{\tau}_1^{p_i}) e^{-j2\pi\hat{\xi}_1^{p_i} t}$ and its discrete form $\hat{s}_1^{i}[k]$, $k = 0, 1, ..., K-1$;

4) Subtract the discrete form of estimation signal of the main target $\hat{s}_1^{i}[k]$ from $x[k]$, obtain the residual echo signal $x_{rest}^{i}[k] = x[k] - \hat{s}_1^{i}[k]$, $k = 0, 1, ..., K-1$ and its energy

$$\mathbf{R}[i] = \Delta t \sum_{k=1}^{K} (x_{rest}^{i}[k])^2. \tag{56}$$

as the residual energy. Searching the minimum value of the matrix $\mathbf{R}$:

$$i_{opt} = \arg\min_{i} \mathbf{R}[i]. \tag{57}$$

As can be seen, there may several optimal exponents that can obtain the minimum value of $\mathbf{R}$, thus to ensure the optimal range resolution we choose the optimal exponent as the smallest one among them, i.e.,

$$p_{opt} = p_0 + \min(i_{opt})(1-p_0)/I;$$

Step 4: Obtain the discrete form of the ECAF of $r(t)$ and $x(t)$ with exponent p_{opt} as $\mathbf{X}_{rx}^{p_{opt}}$; filter $\mathbf{X}_{rx}^{p_{opt}}$ using the wavelet denoising method and obtain the estimated result $w\mathbf{X}_{rx}^{p_{opt}}$; according to the pre-given threshold value H, the discrete form of the JECAF $\theta_{rx}^{p_{opt}}$ is obtained by (52), and then searching the maximum modulus points of $\theta_{rx}^{p_{opt}}$ gives the delay and Doppler shift estimated value of the targets:

$$\hat{m}_l, \hat{n}_l = \arg\max_{m,n} \theta_{rx}^{p_{opt}}[m,n], l = 1, 2, ..., L, \tag{58}$$

the corresponding delay and Doppler shift estimated value of JECAF are $\hat{\tau}_{JECAF}^{l} = \Delta t \times \hat{m}_l, \hat{\xi}_{JECAF}^{l} = \Delta\omega \times \hat{n}_l$;

Step 5: The final estimated value of delay and Doppler shift of the main target are $\hat{\tau}_1 = \hat{\tau}_{JECAF}^{1}, \hat{\xi}_1 = \hat{\xi}_{CAF}$, and the final delay and Doppler shift estimated values of the other targets are $\hat{\tau}_l = \hat{\tau}_{JECAF}^{l}, \hat{\xi}_l = \hat{\xi}_{JECAF}^{l}, l = 2, 3, ..., L$.

IV. Simulation Results and Analysis

In the following simulation experiments, we compare the results of three estimation algorithms based on classical CAF,

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

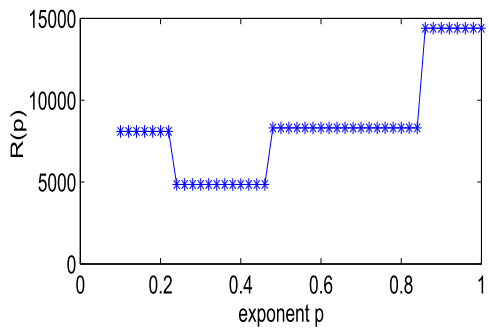


Fig. 7. Curve of residual energy.

ECAF with optimal exponent, and the proposed JECAF with optimal exponent, respectively. We selected an FM signal as the transmitted signal in simulation experiment 1, and a binary coding signal is used as the transmitted signal in simulation experiment 2.

A. Simulation Experiment 1

Initially, we used the FM signal given by (41) as the transmitted signal, with sampling interval $\Delta t = 1/3\ \mu$s, and sampling length $K = 3000\Delta t$. The received signal from the two adjacent targets had true parameter values $\alpha_1 = 1, \tau_1 = 50, \xi_1 = 0, \alpha_2 = 0.8, \tau_2 = 72, \xi = 0$, in the units of the sampling interval. We selected the threshold $H = 0.35$ as the maximum sidelobe level of the normalization of traditional CAF. The SNR loss S_0 was set as 80%, which meant that the output SNR of the EAF was at least 20% of that of the traditional AF. Thus, the lower bound of the exponent is obtained as $p_0 = 0.1$ by solving problem (46).

If input SNR is 0 dB, for each $p \geq p_0$, according to Step 3 in our proposed joint delay and Doppler shift estimation algorithm, we obtained the initial value of α_1 as $\alpha_{initial} = 0.987$ and the delay and Doppler shift estimated values $\hat{\tau}_1^p$ and $\hat{\xi}_1^p$ of the main target by JECAF θ_{rx}^p, and thus we calculated the residual energy $R(p)$ by (56). Fig. 7 gives the curve of the residual energy $R(p)$ as the function of exponent p. We can see that if p is in the interval $[0.24, 0.46]$, the residual energy obtains its minimum value, thus our estimation algorithm gives the optimal exponent as $p_{opt} = 0.24$.

The cut along τ-axis (at all Doppler shift) of the three kinds of cross-ambiguity function, CAF, ECAF with $p = p_{opt}$, and JECAF with $p = p_{opt}$ are given in Fig. 8(a)–(c), respectively. That is, we plot all the range cut of $|\chi_{rx}(\tau, \xi)|$, i.e., $|\chi_{rx}(\tau, \Delta\xi)|, |\chi_{rx}(\tau, 2\Delta\xi)|, ..., |\chi_{rx}(\tau, K\Delta\xi)|$, in Fig. 8(a)–(c); therefore, multiple curves are present.

As can be seen from the traditional CAF we cannot distinguish Target_1 from Target_2; in comparison, from ECAF with $p = p_{opt}$ and JECAF with $p = p_{opt}$ we can clearly distinguish the two targets. At the same time, the estimation errors of the three approaches based on CAF, ECAF and wJECAF are $(2, 0)$, $(1, 0)$ and $(0, 0)$, respectively. This means that for the delay estimation error we have: CAF > ECAF > JECAF; for the Doppler shift estimation error we have: CAF = ECAF = JECAF.

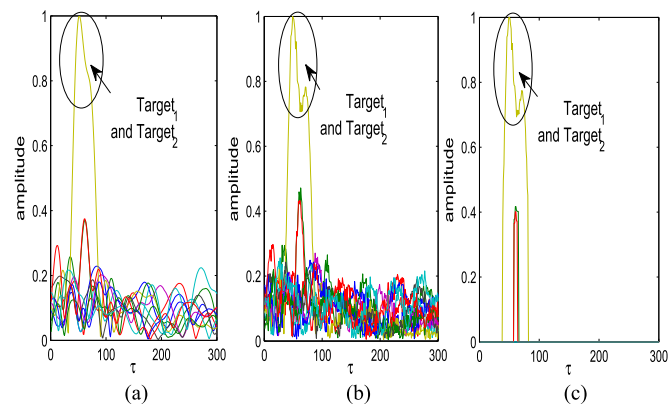


Fig. 8. Cut along τ-axis (at all Doppler shift) of CAF, ECAF with $p = p_{opt}$, and JECAF with $p = p_{opt}$. (a) Cut along τ-axis of CAF. (b) Cut along τ-axis of ECAF with p_{opt}. (c) Cut along τ-axis of JECAF with p_{opt}.

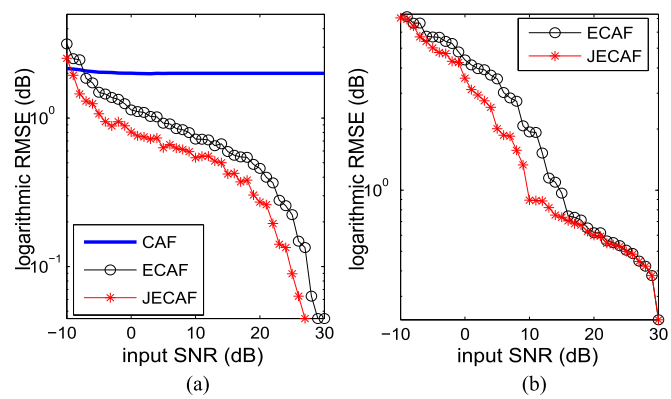


Fig. 9. Delay RMSEs in logarithmic axis of target_1 and target_2. (a) Delay RMSE of target_1. (b) Delay RMSE of target_2.

The delay RMSEs in logarithmic axis of target_1 and target_2 using the three estimation algorithms are given in Fig. 9(a) and (b) respectively. The range of the input SNR is between -10 dB $\sim$30 dB. For each input SNR, we performed 1000 experiments and averaged the results. Fig. 9(a) shows that if input SNR is greater than -10 dB, the delay errors of JECAF and ECAF of target_1 are smaller than that of traditional CAF. Moreover, owing to the wavelet threshold denoising effect, the JECAF has a lower estimation error than ECAF. As the input SNR continues to increase, the noise influence decreases, because CAF has a resolution limit so that its delay estimation error tends to be constant at 2, while the estimation errors of JECAF and ECAF gradually tend toward zero. Moreover, because CAF cannot distinguish the second target and thus cannot estimate Target_2 (as seen in Fig. 8), we only give the estimation errors of ECAF and JECAF in Fig. 9(b). As can be seen, the estimation error of JECAF is lower than that of ECAF because of the wavelet threshold denoising effect; with input SNR increases gradually, both errors decrease and tend to be consistent, until both finally tend toward 0.

In addition, for all input SNR, all three algorithms can calculate accurate Doppler shift estimations of Target_1, mainly because the FM signal has a high velocity resolution; JECAF

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

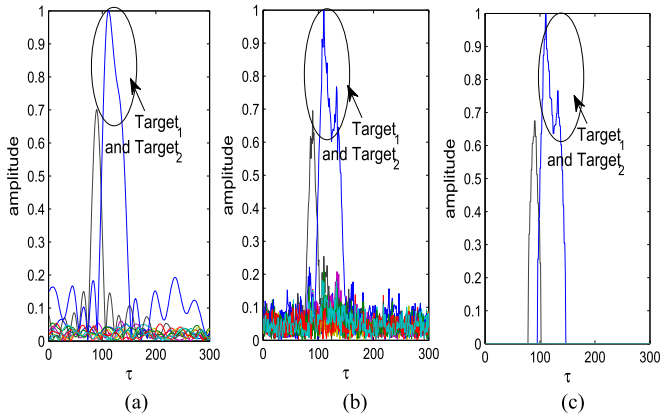


Fig. 10. Cut along τ-axis (at all Doppler shift) of CAF, ECAF with $p = p_{opt}$, and JECAF with $p = p_{opt}$. (a) Cut along τ-axis of CAF. (b) Cut along τ-axis of ECAF with p_{opt}. (c) Cut along τ-axis of JECAF with p_{opt}.

and ECAF are also able to give an accurate Doppler shift estimation of target$_2$.

B. Simulation Experiment 2

Next, we used a phase encoding signal with a 13 bit Barker code [1, 1, 1, 1, 1, −1, −1, 1, 1, −1, 1, −1, 1] as the transmitted signal, with sampling interval $\Delta t = 10^{-2}\ \mu$s and sampling length $K = 8000\Delta t$. In addition, the received echo signal from three adjacent targets has the true parameter values $\alpha_1 = 1, \tau_1 = 110, \xi_1 = 2, \alpha_2 = 0.8, \tau_2 = 133, \xi_2 = 2, \alpha_3 = 0.6, \tau_3 = 90, \xi_3 = 1$, in the units of the sampling interval. We selected the threshold $H = 0.32$. SNR loss S_0 was set at 90%; thus, the lower bound of the exponent is obtained as $p_0 = 0.14$ by solving problem (46).

If input SNR is −5 dB, our estimation algorithm gives the optimal exponent $p_{opt} = 0.14$. The cuts along τ axis (at all Doppler shift) of the three cross ambiguity function, CAF, ECAF with $p = p_{opt}$, and JECAF with $p = p_{opt}$ are given in Fig. 10(a)–(c), respectively. Clearly, traditional CAF cannot distinguish between Target$_1$ and Target$_2$, while ECAF with $p = p_{opt}$ and JECAF with $p = p_{opt}$ can. Concurrently, the estimation errors using CAF, ECAF and JECAF are $(2, 0), (0, 0)$ and $(0, 0)$, respectively. Hence, in this case, for delay estimation error, we have CAF > ECAF = JECAF; for the Doppler shift estimation error we have: CAF = ECAF = JECAF.

The delay RMSE in logarithmic axis of target$_1$ and target$_2$ of the three estimation algorithms based on CAF, ECAF and JECAF are given in Fig. 11(a) and (b), respectively, and the Doppler shift RMSEs in logarithmic axis of target$_1$ and target$_2$ are given in Fig. 11(c) and (d), respectively, in which the range of the input SNR is −20 dB ∼10 dB. For each input SNR, we performed 1000 experiments and averaged the results. As can be seen from Fig. 11(a), for all input SNR, the delay errors of JECAF and ECAF of target$_1$ are smaller than that of traditional CAF. Moreover, because of the wavelet threshold denoising effect, JECAF has lower estimation error than ECAF. As input SNR continues to increase, its delay estimation error tends to be constant at 1 as CAF has a resolution limit, while the estimation

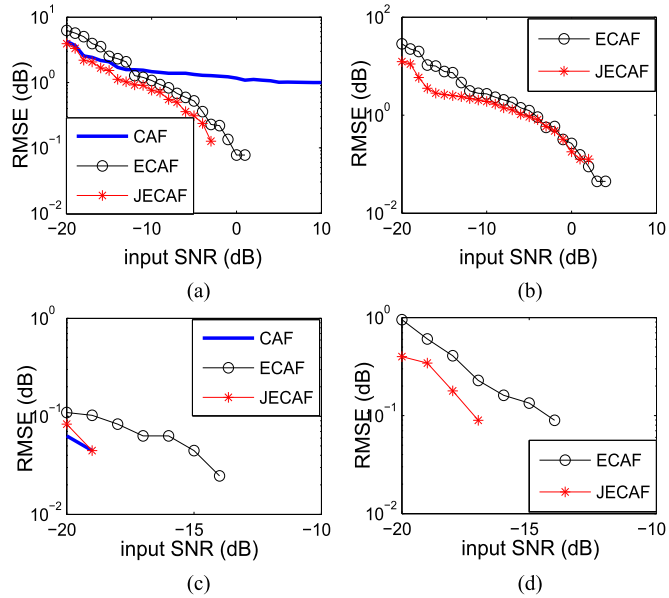


Fig. 11. Delay and Doppler shift RMSEs in logarithmic axis of target$_1$ and target$_2$. (a) Delay RMSE of target$_1$. (b) Delay RMSE of target$_2$. (c) Doppler shift RMSE of target$_1$. (d) Doppler shift RMSE of target$_2$.

errors of JECAF and ECAF gradually tend uniformly to 0 owing to the decreasing noise influence. Moreover, because CAF cannot distinguish the second target, and thus can not achieve the estimation of Target$_2$ (as seen in Fig. 10(a)), we only give the estimation errors of ECAF and JECAF in Fig. 11(b), it can be seen that the estimation error of JECAF is lower than that of ECAF due to the wavelet threshold denoising effect, and as input SNR gradually increases, both errors are decreasing and finally tend toward 0.

Additionally, Fig. 11(c) shows that for all input SNR, the Doppler shift errors are as follows: ECAF > JECAF > CAF. If input SNR is greater than −19 dB, all three algorithms can give accurate Doppler shift estimations of Target$_1$. From Fig. 11 (d), we can see that for input SNR greater than −17 dB, JECAF also give accurate Doppler shift estimations of target$_2$.

V. Conclusion

The ability to identify two targets in the (τ, ξ) plane is limited by the support provided by AF as dictated by the radar uncertainty principle. To solve this, we extended the traditional AF with a new time-frequency analysis tool, that is, the EAF with variable exponent p. For joint delay and Doppler shift estimation, p controls the range resolution, velocity resolution, and output SNR. Theoretical analysis of the properties of EAF shows that the proposed EAF has the following advantages: 1) EAF with $p < 1$ has a higher range resolution than that of AF; 2) according to the optimal exponent selection method, which is adaptive to the received signal, ECAF with optimal exponent achieves the optimal range resolution; and 3) ECAF can transform the loss output SNR to the range resolution and thus make full use of existing resources to achieve a rational allocation of resources.

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

Then we combined traditional CAF and ECAF to form JECAF, which has the best range resolution, velocity resolution, and range and velocity joint resolution when compared with CAF and ECAF. Moreover, we used the wavelet threshold denoising method to combat SNR loss. Therefore, we propose a joint delay and Doppler shift estimation algorithm based on the combination of JECAF and wavelet threshold denoising. The results of the simulation experiments with an FM and with binary coding transmitted signals showed that, for the FM signal, if input SNR > -10 dB, the delay estimation precision of the main target, based on the proposed algorithm, is higher than that of classical AF; meanwhile the Doppler shift estimation precision of the main target based on the proposed algorithm is the same as that of AF. For the binary coding signal, if input SNR > -20 dB, the proposed algorithm attains a higher precision of delay estimation of the main target than classical AF; if input SNR > -19 dB, the Doppler shift estimation of the main target is as accurate as that of AF. However, if there are multiple targets in close proximity to one another, the limited range resolution of AF causes its inability to estimate secondary targets, while JECAF can still estimate parameters with high precision.

Appendix A
Proof of Property 2

Proof: Since,

$$\begin{aligned}\chi_{uu}^{p}(\tau,\xi) &= F[U(f-\xi)U_p^*(f)]\\ &= F[U(f-\xi)U^*(f)|U(f)|^{p-1}],\\ \chi_{vv}^{p}(\tau,\xi) &= F[V(f-\xi)V_p^*(f)]\\ &= F[V(f-\xi)V^*(f)|V(f)|^{p-1}],\end{aligned} \tag{59}$$

then from the uniqueness of the Fourier transform one has, $\chi_{uu}^{p}(\tau,\xi)=\chi_{vv}^{p}(\tau,\xi)$, if and only if for any (τ,ξ) the following equation holds:

$$\frac{U(f-\xi)}{V(f-\xi)}\left[\frac{U(f)}{V(f)}\right]^*\left|\frac{U(f)}{V(f)}\right|^{p-1}=1. \tag{60}$$

If $v(t)=cu(t)$ and $|c|=1$, then (60) clearly establishes. On the contrary, if (60) establishes, in particular, let $f=0,\xi=-\xi$, and suppose

$$c=\left[\frac{U(0)}{V(0)}\right]^*\left|\frac{U(0)}{V(0)}\right|^{p-1}. \tag{61}$$

Then for any ξ, substituting of (61) to (60) yields,

$$cU(\xi)=V(\xi). \tag{62}$$

Therefore, from the uniqueness of the Fourier transform we have $v(t)=cu(t)$. Moveover, let $\xi=0$ in (62) we get $U(0)/V(0)=1/c$, combining it with (61) we finally obtain that $|c|=1$. ■

Appendix B
Proof of Property 3

Proof: Taking the conjugate of (3) one has:

$$[\chi_{rr}^{p}(\tau,\xi)]^*=\int_{-\infty}^{+\infty}R^*(f-\xi)R_p(f)e^{j2\pi f\tau}\mathrm{d}f, \tag{63}$$

thus, multiplying (8) with (63) gives

$$\begin{aligned}|\chi_{rr}^{p}(\tau,\xi)|^2 &= \chi_{rr}^{p}(\tau,\xi)\cdot[\chi_{rr}^{p}(\tau,\xi)]^*\\ &= \iint_{-\infty}^{+\infty} r(t)r_p^*(t+\tau)R^*(f-\xi)R_p(f)\\ &\quad\times\ e^{j2\pi(f\tau+\xi t)}\mathrm{d}t\mathrm{d}f.\end{aligned} \tag{64}$$

Therefore,

$$\begin{aligned}\iint_{-\infty}^{+\infty}|\chi_{rr}^{p}(\tau,\xi)|^2\mathrm{d}\tau\mathrm{d}\xi &= \iiiint_{-\infty}^{+\infty} r(t)r_p^*(t+\tau)\\ &\quad\times\ R^*(f-\xi)R_p(f)e^{j2\pi(f\tau+\xi t)}\mathrm{d}t\mathrm{d}f\mathrm{d}\tau\mathrm{d}\xi.\end{aligned} \tag{65}$$

Using the following relationships of the Fourier transform one has,

$$\int_{-\infty}^{+\infty}y^*(t+\tau)e^{j2\pi f\tau}\mathrm{d}\tau=e^{-j2\pi ft}Y^*(f), \tag{66}$$

$$\int_{-\infty}^{+\infty}Y^*(f-\xi)e^{j2\pi\xi t}\mathrm{d}\xi=e^{j2\pi ft}y^*(t). \tag{67}$$

(65) can be rewritten as:

$$\begin{aligned}&\iint_{-\infty}^{+\infty}|\chi_{rr}^{p}(\tau,\xi)|^2\mathrm{d}\tau\mathrm{d}\xi\\ &= \int_{-\infty}^{+\infty}r(t)r^*(t)\mathrm{d}t\times\int_{-\infty}^{+\infty}R_p(f)R_p^*(f)\mathrm{d}f.\end{aligned} \tag{68}$$

From the Parseval's theorem and using the definition of E_p in (12), (68) can be rewritten as (11) in Property 3. ■

Appendix C
Proof of Property 4

Proof: Multiplying (3) with (63) gives

$$\begin{aligned}|\chi_{rr}^{p}(\tau,\xi)|^2 &= \iint_{-\infty}^{+\infty}[R(\tilde{f}-\xi)]^*R_p(\tilde{f})R(\tilde{v}-\xi)R_p^*(\tilde{v})\\ &\quad\times\ e^{j2\pi(\tilde{v}-\tilde{f})\tau}\mathrm{d}\tilde{v}\mathrm{d}\tilde{f}.\end{aligned}$$

Thus, using the variable substitution $\tilde{v}-\xi=v,\tilde{f}-\xi=f$ we have:

$$\begin{aligned}&\int_{-\infty}^{+\infty}|\chi_{rr}^{p}(\tau,\xi)|^2\mathrm{d}\tau\\ &= \iiint_{-\infty}^{+\infty}R(f)R_p^*(f+\xi)R^*(v)R_p(f+\xi)\\ &\quad\times\ e^{j2\pi(v-f)\tau}\mathrm{d}v\mathrm{d}f\mathrm{d}\tau\\ &= \iiint_{-\infty}^{+\infty}R(f)R_p^*(f+\xi)R^*(v)R_p(v+\xi)\delta(v-f)\mathrm{d}v\mathrm{d}f\\ &= \int_{-\infty}^{+\infty}R(f)R_p^*(f+\xi)R^*(f)R_p(f+\xi)\mathrm{d}f.\end{aligned} \tag{69}$$

Let $\xi=0$ and $p=1$ in (2) we get $\chi_{rr}^{1}(\tau,0)=F[R(f)R^*(f)]$.

Then we have:

$$R(f)R^*(f)=\int_{-\infty}^{+\infty}\chi_{rr}^{1}(\tau,0)e^{j2\pi f\tau}\mathrm{d}\tau,$$

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.

substituting it into (69) yields:

$$\int_{-\infty}^{+\infty} |\chi_{rr}^{p}(\tau,\xi)|^2 \mathrm{d}\tau \tag{70}$$
$$= \int_{-\infty}^{+\infty} \chi_{rr}^{1}(\tau,0) e^{j2\pi\xi\tau} \left[\int_{-\infty}^{+\infty} R_p^*(f) R_p(f) e^{j2\pi f\tau} \mathrm{d}f \right] \mathrm{d}\tau.$$

Using the definition of AF of $r_p(t)$, the above formula can be simplified to the first formula in Property 4. Similarly, the second one can be proved. ■

APPENDIX D
PROOF OF PROPERTY 10

Proof: Since

$$\iint_{-\infty}^{+\infty} |\chi_{rr}^{p}(\tau,\xi)|^2 e^{j2\pi(\tau y-\xi z)} \mathrm{d}\tau \mathrm{d}\xi$$
$$= \iiiint_{-\infty}^{+\infty} r(t) r_p^*(t+\tau) R^*(f-\xi) R_p(f)$$
$$\times e^{j2\pi[(t-z)\xi+(y+f)\tau]} \mathrm{d}t \mathrm{d}f \mathrm{d}\tau \mathrm{d}\xi. \tag{71}$$

Using the properties of the Fourier transform one has:

$$\int_{-\infty}^{+\infty} y^*(t+\tau) e^{j2\pi(y+f)\tau} \mathrm{d}\tau = e^{-j2\pi(y+f)\tau} Y^*(f+y),$$
$$\int_{-\infty}^{+\infty} Y^*(f-\xi) e^{j2\pi(t-z)\xi} \mathrm{d}\xi = e^{j2\pi f(t-z)} y^*(t-z),$$

Introduction of new variables $\rho = t - z, \sigma = f + y$, we get:

$$\iint_{-\infty}^{+\infty} |\chi_{rr}^{p}(\tau,\xi)|^2 e^{j2\pi(\tau y-\xi z)} \mathrm{d}\tau \mathrm{d}\xi$$
$$= \iiiint_{-\infty}^{+\infty} r(\rho+z) R_p(\sigma-y) r_p^*(\rho+z+\tau)$$
$$\times\ e^{-j2\pi\sigma\tau} R^*(\sigma-y-\xi) e^{j2\pi\rho\xi} \mathrm{d}\rho \mathrm{d}\sigma \mathrm{d}\tau \mathrm{d}\xi$$
$$= \int_{-\infty}^{+\infty} R_p^*(\sigma) R_p(\sigma-y) e^{j2\pi\sigma z} \mathrm{d}\sigma$$
$$\times \int_{-\infty}^{+\infty} r^*(\rho) r(\rho+z) e^{j2\pi y\rho} \mathrm{d}\rho$$
$$= \chi_{rr}^{p}(z,y) [\chi_{rr}^{1}(z,y)]^* \tag{72}$$

■

REFERENCES

[1] A. Lanterman, "Radar target resolution probability in a noise-limited environment," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 44, no. 3, pp. 1234–1239, Jul. 2008.

[2] D. W. Ricker, *Echo Signal Processing* (The Springer International Series in Engineering and Computer Science), vol. 725. New York, NY, USA: Springer, 2003.

[3] S. S. Goh, T. N. T. Goodman, and F. Shang, "Joint estimation of delay and doppler shift for band-limited signals," *IEEE Trans. Signal Process.*, vol. 58, no. 9, pp. 4583–4594, Sep. 2010.

[4] T. Derham, S. Doughty, C. Baker, and K. Woodbridge, "Ambiguity functions for spatially coherent and incoherent multistatic radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 46, no. 1, pp. 230–245, Jan. 2010.

[5] B. Boashash and P. Black, "An efficient real-time implementation of the Wigner-Ville distribution," *IEEE Trans. Acoust, Speech, Signal Process.*, vol. 35, no. 11, pp. 1611–1618, Nov. 1997.

[6] S. Barbarossa, "Analysis of multicomponent LFM signals by a combined Wigner-Hough transform," *IEEE Trans. Signal Process.*, vol. 41, no. 6, pp. 1511–1515, Jun. 1995.

[7] D. Mbovilovic and R. N. Bracewell, "Adaptive chirplet respresentation of signals on time-frequency plane," *Electron. Lett.*, vol. 27, pp. 1159–1161, 1991.

[8] O. Akay and G. F. Boudreaux-Bartels, "Fractional Convolution and correlation via operator methods and an application to detection of linear FM signals," *IEEE Trans. Signal Process.*, vol. 49, no. 5, pp. 979–993, May 2001.

[9] S. Barbarossa, A. Scaglione, and G. B. Giannakis, "Product high-order ambiguity function for multicomponent polynomial-phase signal modeling," *IEEE Trans. Signal Process.*, vol. 46, no. 3, pp. 691–708, Mar. 1998.

[10] H. Deng, "Effective CLEAN algorithms for performance-enhanced detection of binary coding radar signals," *IEEE Trans. Signal Process.*, vol. 52, no. 1, pp. 72–78, Jan. 2004.

[11] M. I. Skolnik, *Radar Handbook*, 3rd ed. New York, NY, USA: McGraw-Hill, 2008.

[12] L. Zhang, K. Deng, H.-Q. Wang, and M.-K. Luo, "Exponential filter-based delay estimation algorithm for active systems in the presence of multi-targets, " *IET Radar, Sonar Navig.*, vol. 7, no. 3, pp. 287–294, 2013.

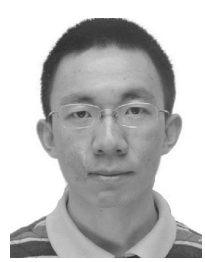
[13] D. L. Donoho, "De-noising by soft-thresholding," *IEEE Trans. Inf. Theory*, vol. 41, no. 3, pp. 613–627, May 1995.

[14] S. Mallat, *A Wavelet Tour of Signal Processing*. Waltham, MA, USA: Academic, 2009.

[15] L. Zhang, K. Deng, H.-Q. Wang, and M.-K. Luo, "Time-delay estimation of multiple targets with wavelet based phase-only matched filter," *Int. J. Electron.*, vol. 101, no. 7, pp. 963–975, 2014.

[16] N. Levanon and E. Mozeson, *Radar Signals*. Hoboken, NJ, USA: Wiley, 2009, pp. 34–52.

Lu Zhang (M'16) received the B.S. degree in applied mathematics in 2007 and the Ph.D. degree in mathematics in 2012, both from the Sichuan University, Chengdu, China. She is an Associate Professor in the College of Mathematics, Sichuan University. Her current research interests include signal processing, statistical physics, and nonlinear dynamics.



Bo Yang received the B.S. degree from Sichuan University, Chengdu, Sichuan, China, in 2010. He is currently working toward the Ph.D. degree in the College of Mathematics, Sichuan University. His current research interests include data analysis, statistical physics, and nonlinear dynamics.

Maokang Luo received the Ph.D. degree from Sichuan University, Chengdu, China, in 1992. He is a Professor and Ph.D. Supervisor in the College of Mathematics, Sichuan University. His main research interests include uncertainty processing and signal processing.

Authorized licensed use limited to: University of Michigan Library. Downloaded on June 09,2022 at 18:50:31 UTC from IEEE Xplore. Restrictions apply.